Soumya Tyagi

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Summary

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, NLP, and Generative AI. Skilled in Python, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit, LangChain, and LLMs. Experienced in building end-to-end AI solutions including data preprocessing, model training, evaluation, semantic matching, and AI automation. Strong foundation in DSA and core computer science concepts.

Education

Vellore Institute of Technology 2023 – 2027

B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)

CGPA: **8.43/10**

Technical Skills

Programming: Python, SQL, Java (DSA)

AI & Machine Learning: Machine Learning, Deep Learning, NLP, LLMs, Generative AI, RAG, Embeddings, Hugging Face

Frameworks & Backend: LangChain, LangGraph, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit

Cloud & Tools: AWS, Docker

Core CS: OOP, DSA, DBMS

Projects

RepoMind AI — Agentic RAG GitHub Repository Intelligence | *Python · LangChain · ChromaDB · Sentence Transformers · Groq LLaMA · GitHub API* August 2026

- Engineered an **AI-powered GitHub codebase assistant** using LangChain, ChromaDB, and Groq, building 4+ specialized tools to automate multi-step repository search and code analysis.
- Improved RAG retrieval accuracy by 25% over the baseline pipeline by integrating query expansion, multi-query retrieval, cross-encoder re-ranking, and context compression.
- Expanded code search coverage by 3× for complex queries by replacing single-step retrieval with agentic tool routing across files, languages, and repo structures.

AI Production Incident Investigator | *Python · LangGraph · LangChain · Groq · RAG · ChromaDB · FastAPI · Streamlit* Oct 2026

- Built an **AI-powered Production Incident Investigator** with Python, LangChain, LangGraph, RAG, and LLMs to analyze production incidents using logs, system documentation, and past incident data.
- Developed a 6-stage LangGraph workflow covering incident analysis, hypothesis generation, evidence retrieval, evaluation, and re-investigation, using ChromaDB and Sentence Transformers for semantic search.
- Built FastAPI and a Streamlit interface that generate structured incident reports with root cause, confidence scores, supporting/conflicting evidence, and actionable remediation steps.

Extracurricular Activities

Johns Hopkins Health Hackathon

- Contributed to an AI-powered nutrition recommendation system by developing the responsive frontend and integrating it with backend APIs for food analysis and personalized dietary insights.

Adobe University Hackathon 2026

- Participated in the Adobe University Hackathon 2026, collaborating with a team to develop and present a technology solution within a competitive hackathon environment.

Inter-College Basketball Tournament

- Secured **1st Place** with the college basketball team, demonstrating teamwork, leadership, discipline, and competitive spirit.